

Academic Year	11	Calendar Year	2021
Course	Ext	Name of task/exam	Yearly

Section I

1. $|x-3| > 4$

$$\begin{array}{ll} x-3 > 4 & -(x-3) > 4 \\ x > 7 & x-3 < -4 \\ & x < -1 \end{array}$$

$\therefore x < -1, x > 7$ (B)

2. $y = |x|$ fails the horizontal line test. (D)

3. zeroes at 1 and -2

Equation is of the form

$y = a(x-1)^2(x+2)^3$

Two options: A or B.

a must be negative
so (B).

4. $\frac{7!}{2!3!} = 420$ (A)
 $\begin{array}{cc} \uparrow & \uparrow \\ 2D's & 3E's \end{array}$

5. Eliminate A and D as these graphs need to be above/on the x-axis.

Eliminate B as the section above/on the x-axis of $f(x)$ should remain the same $\therefore$ (C)Section II

6. a) $p(x) = x^3 - 2x^2 + kx + 24$

$$\begin{aligned} \text{(i)} \quad \alpha + \beta + \gamma &= \frac{-b}{a} \\ &= \frac{-(-2)}{1} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \alpha\beta\gamma &= \frac{-d}{a} \\ &= \frac{-24}{1} \\ &= -24 \end{aligned}$$

(iii) let the roots be $\alpha, -\alpha, \beta$

$\alpha + -\alpha + \beta = 2$

$\beta = 2$

 $\therefore$ one of the roots is 2.

$p(2) = 0$

$0 = 2^3 - 2(2)^2 + k(2) + 24$

$0 = 2k + 24$

$-24 = 2k$

$\therefore k = -12$

b) $\frac{3}{x-2} \cdot (x-2)^2 \leq 1 \cdot (x-2)^2$

$3(x-2) \leq (x-2)^2$

$0 \leq (x-2)^2 - 3(x-2)$

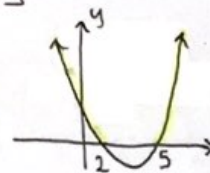
$0 \leq (x-2)[x-2-3]$

$0 \leq (x-2)(x-5)$

$(x-2)(x-5) \geq 0$

$x < 2, x \geq 5$

$(x \neq 2)$



Solutions for exams and assessment tasks

Academic Year	11	Calendar Year	2021
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6. cont.

c) (i) $5+7=12$ students in total.

$${}^{12}C_7 = 792$$

(ii) 4 high school students

This means 3 primary school students.

$${}^7C_4 \cdot {}^5C_3 = 350$$

Probability is $\frac{350}{792} = \frac{175}{396}$

d) $x = t + 4$ $y = 2 + t^2$

(i) $t = x - 4$

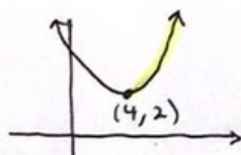
sub $t = x - 4$ into $y = 2 + t^2$

$$y = 2 + (x - 4)^2$$

$$y = (x - 4)^2 + 2$$

$$f(x) = (x - 4)^2 + 2$$

(ii)

Largest possible domain forwhen $f(x)$ is monotonically increasing: $x \geq 4$

(iii) $f: y = (x - 4)^2 + 2$

Inverse relation:

$$x = (y - 4)^2 + 2$$

$$x - 2 = (y - 4)^2$$

$$y - 4 = \pm \sqrt{x - 2}$$

$$y = 4 \pm \sqrt{x - 2}$$

$$\therefore f^{-1}(x) = 4 + \sqrt{x - 2} \text{ as } y \geq 4$$

(Domain of $f(x)$ is $x \geq 4$ so range of $f^{-1}(x)$ is $y \geq 4$).

6. cont.

e) $x = \frac{t^3}{3} - \frac{5t^2}{2} + 6t + 4$

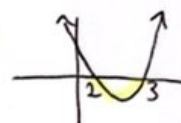
$$\frac{dx}{dt} = \frac{3t^2}{3} - \frac{10t}{2} + 6$$

$$= t^2 - 5t + 6$$

$$t^2 - 5t + 6 < 0$$

$$(t - 3)(t - 2) < 0$$

$$\therefore 2 < t < 3$$



7. a) $P(x) = x^3 - 3x^2 + kx - 5$

$$P(-2) = 7$$

$$7 = (-2)^3 - 3(-2)^2 + k(-2) - 5$$

$$7 = -2k - 25$$

$$32 = -2k$$

$$k = -16$$

b) $y = 3 \cos^{-1} 2x$

D: $-1 \leq 2x \leq 1$

$$-\frac{1}{2} \leq x \leq \frac{1}{2}$$

R: $0 \leq \cos^{-1} 2x \leq \pi$

$$0 \leq 3 \cos^{-1} 2x \leq 3\pi$$

$$0 \leq y \leq 3\pi$$

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7. cont.

$$c) {}^{n+1}C_r \div {}^nC_{r-1} = \frac{n+1}{r}$$

$$\text{LHS} = {}^{n+1}C_r \div {}^nC_{r-1}$$

$$= \frac{(n+1)!}{r!(n+1-r)!} \div \frac{n!}{(r-1)!(n-(r-1))!}$$

$$= \frac{(n+1)\cancel{n!}}{r(r-1)!(n+1-r)!} \times \frac{(r-1)!(n-r+1)!}{\cancel{n!}}$$

$$= \frac{n+1}{r}$$

$$= \text{RHS}$$

$$d) (i) (9-1)! = 8! = 40320$$

(ii) Total arrangements - Number of arrangements they do sit together

$$= 8! - (2 \times 7!)$$

$$= 40320 - 10080$$

$$= 30240$$

OR

$$\underline{1} \times \underline{6} \times \underline{7!} = 30240$$

1 seat Steve or Danny.

6 the other has 6 seats he can select where they are not sitting next to each other.

7! arrangements for the remaining 7 people.

$$e) (i) \tan 2A$$

$$= \tan(A+A)$$

$$= \frac{\tan A + \tan A}{1 - \tan A \tan A}$$

$$= \frac{2 \tan A}{1 - \tan^2 A}$$

$$(ii) \tan 2A = \frac{2 \tan A}{1 - \tan^2 A} \quad \text{from (i)}$$

$$\tan 2(67.5^\circ) = \frac{2 \tan 67.5^\circ}{1 - \tan^2 67.5^\circ}$$

$$\tan 135^\circ = \frac{2 \tan 67.5^\circ}{1 - \tan^2 67.5^\circ}$$

$$-1 = \frac{2 \tan 67.5^\circ}{1 - \tan^2 67.5^\circ}$$

$$\text{let } u = \tan 67.5^\circ$$

$$-1 = \frac{2u}{1-u^2}$$

$$-(1-u^2) = 2u$$

$$-1 + u^2 = 2u$$

$$u^2 - 2u - 1 = 0$$

$$u^2 - 2u + 1^2 = 1 + 1^2$$

$$(u-1)^2 = 2$$

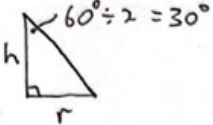
$$u-1 = \pm \sqrt{2}$$

$$u = 1 \pm \sqrt{2}$$

$$\tan 67.5^\circ = 1 + \sqrt{2} \quad \text{as } \tan 67.5^\circ > 0$$

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7. cont.

f) (i) 

$$\tan 30^\circ = \frac{r}{h}$$

$$h \cdot \frac{1}{\sqrt{3}} = \frac{r}{h} \cdot h$$

$$\frac{h}{\sqrt{3}} = r$$

$$\therefore r = \frac{h}{\sqrt{3}}$$

(ii) $V = \frac{1}{3} \pi r^2 h$

$$= \frac{1}{3} \pi \left(\frac{h}{\sqrt{3}} \right)^2 h$$

$$= \frac{1}{3} \pi \cdot \frac{h^2}{3} \cdot h$$

$$\therefore V = \frac{\pi h^3}{9}$$

(iii) $\frac{dh}{dt} = ?$ when $h = 3$ m

$$\frac{dV}{dt} = 0.5 \text{ m}^3/\text{s}$$

$$V = \frac{\pi h^3}{9}$$

$$\frac{dV}{dh} = \frac{3\pi h^2}{9}$$

$$= \frac{\pi h^2}{3}$$

when $h = 3$, $\frac{dV}{dh} = \frac{\pi \times 3^2}{3} = 3\pi$

7. f) (iii) cont.

$$\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV}$$

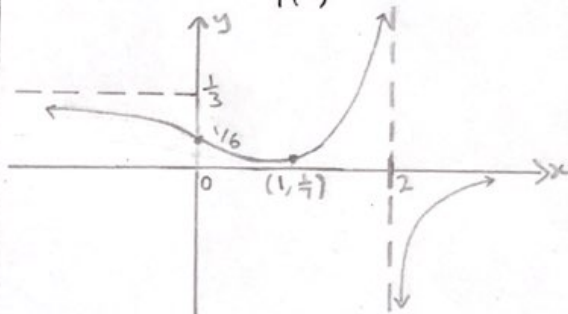
$$= 0.5 \times \frac{1}{3\pi}$$

$$= \frac{1}{2} \times \frac{1}{3\pi}$$

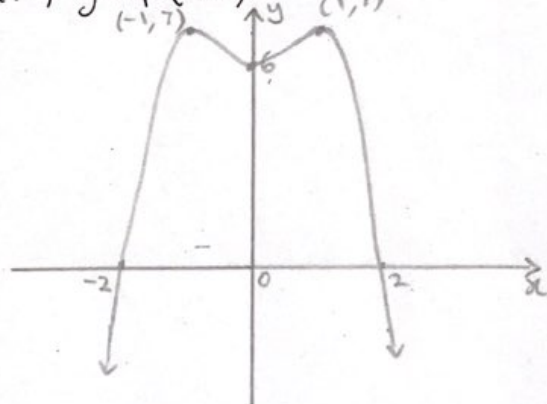
$$= \frac{1}{6\pi}$$

$\therefore$ height of pile is increasing at a rate of $\frac{1}{6\pi}$ m/s.

8. a) (i) $y = \frac{1}{f(x)}$



(ii) $y = f(1 \times 1)$



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8. cont.

b) (i) $T = A + Be^{kt}$

$$\frac{dT}{dt} = k \cdot Be^{kt}, \quad Be^{kt} = T - A$$

$$= k(T - A)$$

(ii) $A = 25^\circ\text{C}$ (surrounding air temperature)

$t = 0, T = 5^\circ\text{C}$

$$5 = 25 + Be^0$$

$$5 = 25 + B$$

$$B = -20$$

So $T = 25 - 20e^{kt}$

$t = 20, T = 15^\circ\text{C}$

$$15 = 25 - 20e^{k(20)}$$

$$-10 = -20e^{20k}$$

$$e^{20k} = \frac{1}{2}$$

$$20k = \ln\left(\frac{1}{2}\right)$$

$$k = \frac{1}{20} \ln\left(\frac{1}{2}\right)$$

a further 50 minutes,

$$t = 20 + 50 = 70 \text{ minutes}$$

$$t = 70, T = 25 - 20e^{\left(\frac{1}{20} \ln\left(\frac{1}{2}\right)\right)(70)}$$

$$= 23.2322\dots$$

$$\doteq 23^\circ\text{C}$$

8. cont.

c) $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ (property of inverse trig functions).

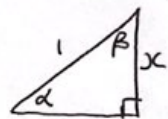
Let $\alpha = \sin^{-1}x$ and $\beta = \cos^{-1}x$.

From the property above, α and β are complementary angles.

From above,

$$\alpha = \sin^{-1}x \quad \text{and} \quad \beta = \cos^{-1}x$$

$$\sin \alpha = x \quad \cos \beta = x$$



$\sqrt{1-x^2}$ (Pythagoras' theorem)

$$\text{LHS} = \sin(\sin^{-1}x - \cos^{-1}x)$$

$$= \sin(\alpha - \beta)$$

$$= \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= x \cdot x - (\sqrt{1-x^2} \cdot \sqrt{1-x^2})$$

$$= x^2 - (1-x^2)$$

$$= x^2 - 1 + x^2$$

$$= 2x^2 - 1$$

$$= \text{RHS}$$

Solutions for exams and assessment tasks

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8. cont.

d) (i) $t = \tan \frac{\theta}{2}$

$$\sin \theta = \frac{2t}{1+t^2} \quad \cos \theta = \frac{1-t^2}{1+t^2}$$

(reference sheet)

$$\text{LHS} = \frac{1}{2} (3 \cos \theta + 4 \sin \theta + 5)$$

$$= \frac{1}{2} \left(3 \left(\frac{1-t^2}{1+t^2} \right) + 4 \left(\frac{2t}{1+t^2} \right) + 5 \right)$$

$$= \frac{1}{2} \left[\frac{3(1-t^2) + 4(2t) + 5(1+t^2)}{1+t^2} \right]$$

$$= \frac{(3 - 3t^2 + 8t + 5 + 5t^2)}{2(1+t^2)}$$

$$= \frac{(2t^2 + 8t + 8)}{2(1+t^2)}$$

$$= \frac{2(t^2 + 4t + 4)}{2(1+t^2)}$$

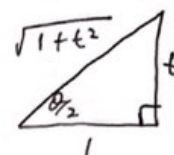
$$= \frac{(t+2)^2}{1+t^2}$$

$$t = \tan \frac{\theta}{2}$$

$$\sin \frac{\theta}{2} = \frac{t}{\sqrt{1+t^2}}$$

$$\cos \frac{\theta}{2} = \frac{1}{\sqrt{1+t^2}}$$

(Pythagoras' Theorem)



$$\text{RHS} = \left(\sin \frac{\theta}{2} + 2 \cos \frac{\theta}{2} \right)^2$$

$$= \left(\frac{t}{\sqrt{1+t^2}} + 2 \left(\frac{1}{\sqrt{1+t^2}} \right) \right)^2$$

$$= \left(\frac{t+2}{\sqrt{1+t^2}} \right)^2$$

$$= \frac{(t+2)^2}{(\sqrt{1+t^2})^2}$$

$$= \frac{(t+2)^2}{1+t^2}$$

$$= \text{LHS}$$

d) (ii) $\theta = \frac{\pi}{3}$

$$\frac{1}{2} (3 \cos \frac{\pi}{3} + 4 \sin \frac{\pi}{3} + 5) = \left(\sin \left(\frac{\pi}{2} \right) + 2 \cos \left(\frac{\pi}{2} \right) \right)^2$$

$$\frac{1}{2} \left(3 \left(\frac{1}{2} \right) + 4 \left(\frac{\sqrt{3}}{2} \right) + 5 \right) = \left(\sin \frac{\pi}{6} + 2 \cos \frac{\pi}{6} \right)^2$$

$$\frac{1}{2} \left(\frac{13}{2} + 2\sqrt{3} \right) = \left(\frac{1}{2} + 2 \left(\frac{\sqrt{3}}{2} \right) \right)^2$$

$$\frac{13 + 4\sqrt{3}}{2} = 2 \left(\frac{1}{2} + \sqrt{3} \right)^2$$

$$13 + 4\sqrt{3} = 4 \left(\frac{1 + 2\sqrt{3}}{2} \right)^2$$

$$13 + 4\sqrt{3} = \frac{4(1 + 2\sqrt{3})^2}{2^2}$$

$$13 + 4\sqrt{3} = (1 + 2\sqrt{3})^2$$

$\therefore$ the two square roots of $13 + 4\sqrt{3}$ are

$$\pm \sqrt{(1 + 2\sqrt{3})^2}$$

$$= \pm (1 + 2\sqrt{3})$$



2021 Year 11 Yearly Examination
Weighting 40%

Mathematics Extension 1

**General
Instructions**

- Reading time – 10 minutes
- Working time – 1.5 hours
- Write using black pen
- Calculators approved by NESA may be used
- The HSC reference sheet may be used
- For questions in Section II, show relevant mathematical reasoning and/or calculation
- **This is an open book examination.**

**Total
marks:
50**

Section I – 5 marks

- Attempt Questions 1–5
- Allow about 10 minutes for this section

Section II – 45 marks

- Attempt Questions 6–8
- Allow about 1 hour and 20 minutes for this section

Marks awarded

Section I	/5
Section II	/45
Total	/50

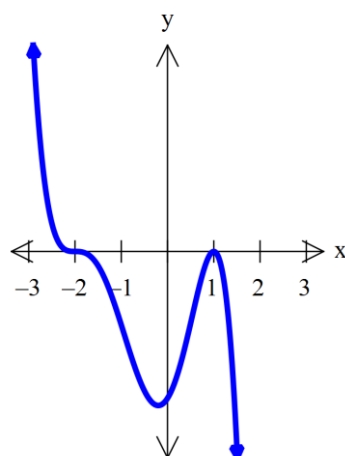
Section I: Multiple Choice (5 marks)**Marks****Start a new writing booklet.**

Attempt questions 1 – 5.

Write the correct alternative (A, B, C or D) for each question.

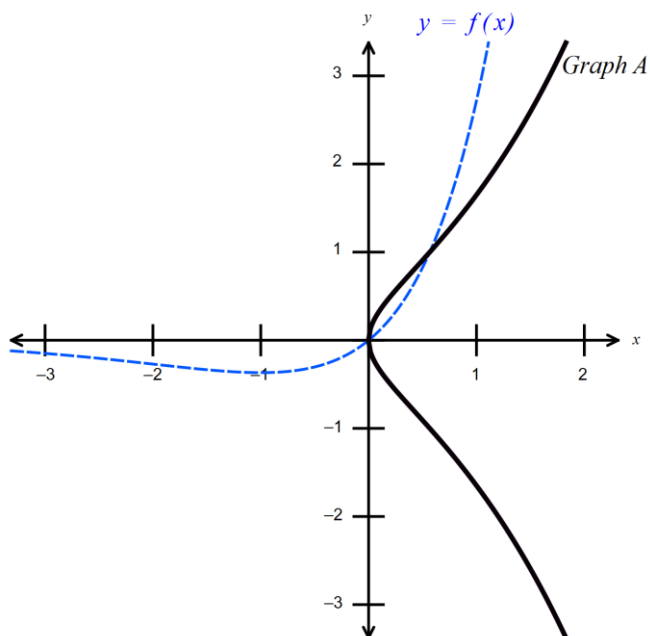
Allow about 10 minutes for this section.

1. Which of the following is the solution to $|x - 3| > 4$? **1**
- (A) $-1 < x < 7$
- (B) $x < -1$ or $x > 7$
- (C) $x < -7$ or $x > 7$
- (D) $x > 1$ or $x < -1$
2. Which one of these functions has an inverse relation that is not a function? **1**
- (A) $y = x^3$
- (B) $y = \sqrt{x}$
- (C) $y = e^x$
- (D) $y = |x|$
3. Which equation best represents the following graph? **1**



- (A) $y = (x - 1)^2(x + 2)^3$
- (B) $y = -(x - 1)^2(x + 2)^3$
- (C) $y = (x + 1)^2(x - 2)^3$
- (D) $y = -(x + 1)^2(x - 2)^3$

4. How many seven-letter arrangements can be made using the letters of the word DELETED? 1
- (A) 420
(B) 840
(C) 2520
(D) 5040
5. The graph of $y = f(x)$ is shown on the diagram below in blue. 1
- Which equation best represents the transformation that produced *Graph A* shown below?



- (A) $y = |f(x)|$
(B) $|y| = f(x)$
(C) $y^2 = f(x)$
(D) $y = \sqrt{f(x)}$

END OF SECTION I

Section II (45 marks)

Start a new writing booklet for each question.

Attempt questions 6 – 8.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Allow about 1 hour and 20 minutes for this section.

Question 6 (15 marks) Start a new writing booklet.

- a) The polynomial $P(x) = x^3 - 2x^2 + kx + 24$ has roots α , β and γ .
- (i) Find the value of $\alpha + \beta + \gamma$. 1
- (ii) Find the value of $\alpha\beta\gamma$. 1
- (iii) It is known that two of the roots are equal in magnitude but opposite in sign. Find the value of k . 2
- b) Solve $\frac{3}{x-2} \leq 1$. 3
- c) A seven-person committee is to be chosen at random from 5 primary school students and 7 high school students.
- (i) In how many ways can this committee be chosen? 1
- (ii) What is the probability that the committee will consist of exactly 4 high school students? 1

Question 6 continues on the next page

Question 6 continued

d) A parabola is defined by the parametric equations $x = t + 4$ and $y = 2 + t^2$.

(i) Show that the Cartesian equation of the parabola is $y = (x - 4)^2 + 2$. **1**

Rewrite the equation of the parabola as $f(x) = (x - 4)^2 + 2$.

(ii) Write the largest possible restricted domain of $f(x)$ such that $f(x)$ is monotonically increasing. **1**

(iii) Find the equation of the inverse function $f^{-1}(x)$ on the domain found in part (ii). Justify your answer. **2**

e) The displacement x metres from the origin at time t seconds of a particle travelling in a straight line is given by $x = \frac{t^3}{3} - \frac{5t^2}{2} + 6t + 4$. **2**
For what values of t is the velocity of the particle negative?

END OF QUESTION 6

Question 7 (16 marks)**Start a new writing booklet.**

- a) Let $P(x) = x^3 - 3x^2 + kx - 5$, where k is a rational number. 2
When $P(x)$ is divided by $(x + 2)$, the remainder is 7. Find k .
- b) State the domain and range of $y = 3 \cos^{-1}(2x)$. 2
- c) Show that ${}^{n+1}C_r \div {}^nC_{r-1} = \frac{n+1}{r}$. 2
- d) Nine people are to be seated at a round table.
- (i) How many seating arrangements are possible? 1
- (ii) Two people, Steve and Danny, refuse to sit next to each other. 2
How many seating arrangements are then possible?
- e) (i) Write $\tan 2A$ in terms of $\tan A$. 1
- (ii) Hence find the exact value of $\tan \left(67\frac{1}{2}\right)^\circ$. 2

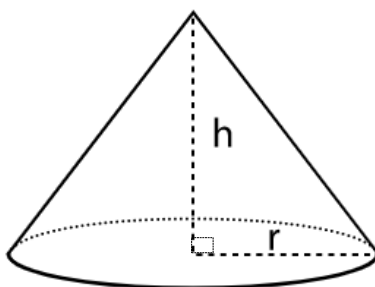
Question 7 continues on the next page

Question 7 continued

- f) Grain is poured at a constant rate of 0.5 cubic metres per second. It forms a conical pile, with the angle at the apex of the cone equal to 60° .

The height of the pile is h metres, and the radius of the base is r metres.

The volume of a cone is given by $V = \frac{1}{3}\pi r^2 h$, where V is the volume, r is the radius and h is the height.

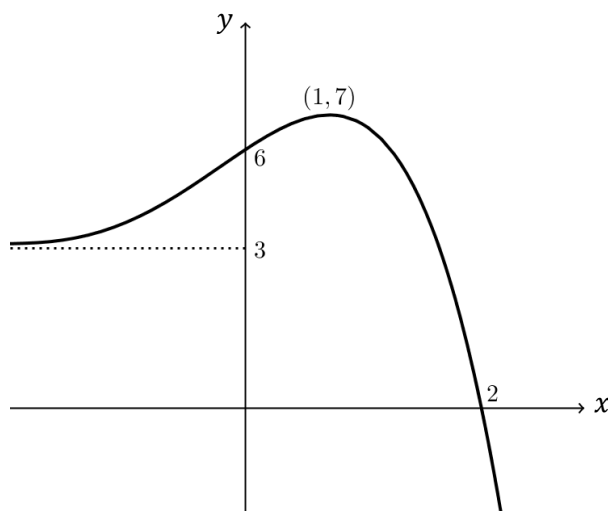


- | | | |
|-------|---|---|
| (i) | Show that $r = \frac{h}{\sqrt{3}}$. | 1 |
| (ii) | Show that V , the volume of the pile, is given by $V = \frac{\pi h^3}{9}$. | 1 |
| (iii) | Hence find the rate at which the height of the pile is increasing when the height of the pile is 3 metres. Leave your answer in exact form. | 2 |

END OF QUESTION 7

Question 8 (14 marks)[Start a new writing booklet.](#)

- a) The following diagram shows the graph of $y = f(x)$.



On separate diagrams, draw graphs of each of the following, carefully showing any asymptotes and all key points.

(i) $y = \frac{1}{f(x)}$ 2

(ii) $y = f(|x|)$ 2

- b) The rate at which a body warms in air is proportional to the difference between its temperature, T , and the constant temperature, A , of the surrounding air.

This rate can be expressed by the differential equation $\frac{dT}{dt} = k(T - A)$, where t is the time in minutes and k is a constant.

- (i) Show that $T = A + Be^{kt}$, where B is a constant, is a solution of the differential equation. 1

- (ii) An object warms from 5°C to 15°C in 20 minutes. The temperature of the surrounding air is 25°C . 3

Find the temperature of the object after a further 50 minutes have elapsed. Give your answer to the nearest degree.

Question 8 continues on the next page

Question 8 continued

- c) Given that $\sin^{-1} x$ and $\cos^{-1} x$ are acute, show that 2

$$\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1$$

- d) (i) Use the substitution $t = \tan \frac{\theta}{2}$ to prove the identity 3

$$\frac{1}{2}(3 \cos \theta + 4 \sin \theta + 5) = \left(\sin \frac{\theta}{2} + 2 \cos \frac{\theta}{2} \right)^2$$

- (ii) Hence use the substitution $\theta = \frac{\pi}{3}$ to find the two square roots of 1
 $(13 + 4\sqrt{3})$. Leave your answers in exact form.

END OF EXAM